

How This Project Works: A Catch-Up Explainer

Cross-Generator Generalization in Deepfake Detection | IE7374: Generative AI

1. The core idea

A deepfake detector is a model that looks at a face and decides real or fake. It learns by example: we show it many labeled real and fake faces, and it picks up the small artifacts that a fake-making tool leaves behind, such as odd blending along the jaw or warped skin texture. The problem we study is that a detector tends to learn the fingerprints of the specific tools in its training set. Show it a fake from a tool it has never seen, and it often misses it, much like a bank teller trained to catch one counterfeiter's bills who is then fooled by a different counterfeiter. Our project measures exactly how much accuracy is lost when the fake comes from a tool the detector was never trained on.

2. The question we actually answer

Not "can the detector spot fakes," which is easy on familiar fakes, but "can it spot fakes made by a method it was never trained on." That is the honest and useful question, and it is the one most projects skip by testing only on the kind of fake the model already knows.

3. How the experiment works

FaceForensics++ gives us four different fake-making methods (DeepFakes, Face2Face, FaceSwap, NeuralTextures). To test generalization, we hide one method during training and use it only for testing.

Worked example (one of eight runs)

Train a model on DeepFakes, Face2Face, and FaceSwap fakes, plus real videos. Then test it only on NeuralTextures fakes, which it has never seen. If it still catches them, it generalized. If its accuracy drops sharply, it did not.

We rotate which method is hidden (four choices) and repeat for both models (EfficientNet and XceptionNet). Four hidden methods times two models is why there are eight training runs.

4. How to read the result

Each run gives two numbers: accuracy on the methods it trained on (usually high, around 0.96) and accuracy on the hidden method (often much lower). The gap between them is the finding. We collect all of these into one table, called a transfer matrix, so we can see which methods transfer well and which do not. Accuracy is measured with AUC: think of it as the chance the model rates a random fake as more suspicious than a random real face. 100 means always correct, 50 means a coin flip.

5. One rule that keeps it honest

We make sure the same person never appears in both training and testing. Otherwise the model can cheat by recognizing the face instead of detecting the manipulation, which would make the results look better than they really are. FaceForensics++ provides an official split that already enforces this.

6. What you would actually do

Pick one or two of the eight runs: you would fine-tune a model on three methods, evaluate it on the held-out fourth, and record its AUC. Or own a pipeline stage: preparing the data, cropping the faces, running the models, or writing up results. All eight runs use the same settings so the numbers are comparable, so your part drops straight into the shared table. See the sign-up sheet and the meeting agenda for the runs and the open decisions.